(FP2) 3. Differentiation 作业2

NO. 1

Find the Maclaurin's series for $\tan(x + \frac{1}{4}\pi)$ up to and including the term in x^2. [5]

A level FURTHER MATHEMATICS 2020/Winter/Paper2_2/Q1

Mark scheme

Question	Answer	Marks	Guidance
1	$f'(x) = \sec^2\left(x + \frac{1}{4}\pi\right)$	**B1**	Finds first derivative.
	$f''(x) = 2\sec^2\left(x + \frac{1}{4}\pi\right)\tan\left(x + \frac{1}{4}\pi\right)$	**B1**	Finds second derivative.
	$f(0) = 1$ $f'(0) = 2$ $f''(0) = 4$	**M1**	Evaluates derivatives at zero.
	$\tan\left(x + \frac{1}{4}\pi\right) = 1 + 2x + 2x^2$	**M1 A1**	
		5	

Explanation

1.

Maclaurin's series are given by $y(0) + y'(0)x + y''(0)\frac{x^2}{2!} + \cdots$

$y = tan\left(x + \frac{\pi}{4}\right)$

$\frac{dy}{dx} = sec^2\left(x + \frac{\pi}{4}\right)$ **B1**

$\frac{d^2y}{dx^2} = 2\,sec\left(x + \frac{\pi}{4}\right) \cdot tan\left(x + \frac{\pi}{4}\right) sec\left(x + \frac{\pi}{4}\right)$

$= 2\,tan\left(x + \frac{\pi}{4}\right) sec^2\left(x + \frac{\pi}{4}\right)$ **B1**

When x=0

$y = 1,\ \frac{dy}{dx} = \frac{1}{cos^2\left(\frac{\pi}{4}\right)} = 2,\ \frac{d^2y}{dx^2} = 2 \cdot 1 \cdot 2 = 4$ **M1**

$y(0) + y'(0)x + y''(0)\frac{x^2}{2!} = 1 + 2x + 2x^2$ **M1 A1**

Comment

This question was very well answered with most candidates accurately finding the Maclaurin's series, but a few did not divide f''(0) by 2.

NO. 2

a. Starting from the definitions of tanh and sech in terms of exponentials, prove that $1 - \tanh^2\theta = \operatorname{sech}^2\theta$. [3]

The variables x and y are such that $\tanh y = \cos\left(x + \frac{1}{4}\pi\right)$, for $-\frac{1}{4}\pi < x < \frac{3}{4}\pi$.

b. By differentiating the equation $\tanh y = \cos\left(x + \frac{1}{4}\pi\right)$ with respect to x, show that $\frac{\mathrm{d}y}{\mathrm{d}x} = -\operatorname{cosec}\left(x + \frac{1}{4}\pi\right)$. [4]

c. Hence find the first three terms in the Maclaurin's series for $\tanh^{-1}\left(\cos\left(x + \frac{1}{4}\pi\right)\right)$ in the form $\frac{1}{2}\ln a + bx + cx^2$, giving the exact values of the constants a, b and c. [5]

A level FURTHER MATHEMATICS 2020/Summer/Paper2_3/Q6

Mark scheme

Question	Answer	Marks
a	$\tanh\theta = \dfrac{e^{\theta}-e^{-\theta}}{e^{\theta}+e^{-\theta}} \quad \operatorname{sech}\theta = \dfrac{2}{e^{\theta}+e^{-\theta}}$	B1
	$1-\left(\dfrac{e^{\theta}-e^{-\theta}}{e^{\theta}+e^{-\theta}}\right)^2 = \dfrac{\left(e^{\theta}+e^{-\theta}\right)^2-\left(e^{\theta}-e^{-\theta}\right)^2}{\left(e^{\theta}+e^{-\theta}\right)^2} = \dfrac{4}{\left(e^{\theta}+e^{-\theta}\right)^2}$	**M1**
	$= \operatorname{sech}^2\theta$	**A1**
		3
b	$\tanh y = \cos(x+\frac{1}{4}\pi) \quad \Rightarrow \quad \operatorname{sech}^2 y \dfrac{dy}{dx} = -\sin(x+\frac{1}{4}\pi)$	**M1 A1**
	$\left(1-\cos^2(x+\frac{1}{4}\pi)\right)\dfrac{dy}{dx} = -\sin(x+\frac{1}{4}\pi)$	**M1**
	$\dfrac{dy}{dx} = -\dfrac{\sin(x+\frac{1}{4}\pi)}{\sin^2(x+\frac{1}{4}\pi)} = -\operatorname{cosec}(x+\frac{1}{4}\pi)$	**A1**
		4
c	$\dfrac{d^2y}{dx^2} = \cot(x+\frac{1}{4}\pi)\operatorname{cosec}(x+\frac{1}{4}\pi)$	**B1**
	$y'(0) = -\operatorname{cosec}(\frac{1}{4}\pi) \quad y''(0) = \cot(\frac{1}{4}\pi)\operatorname{cosec}(\frac{1}{4}\pi)$	**M1**
	$y(0) = \tanh^{-1}\left(\frac{1}{2}\sqrt{2}\right) = \frac{1}{2}\ln\left(\dfrac{2+\sqrt{2}}{2-\sqrt{2}}\right)$	**M1**
	$y = \frac{1}{2}\ln(3+2\sqrt{2}) - x\sqrt{2} + \frac{1}{2}x^2\sqrt{2}$	**M1 A1**

		5

Explanation

a.

$$tanh\,\theta = \frac{sinh\,\theta}{cosh\,\theta} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$sech\,\theta = \frac{1}{cosh\,\theta} = \frac{2}{e^x + e^{-x}}$$ **B1**

$$1 - tanh^2\,\theta = 1 - \left(\frac{e^x - e^{-x}}{e^x + e^{-x}}\right)^2 = \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{(e^x + e^{-x})^2}$$

$$= \frac{4}{(e^x + e^{-x})^2} = \left(\frac{2}{e^x + e^{-x}}\right)^2 = sech^2\,\theta$$ **M1A1**

b)

differentiate respect to x: $tanh\,y = cos\left(x + \frac{\pi}{4}\right)$

$$sech^2\,y\frac{dy}{dx} = -sin\left(x + \frac{\pi}{4}\right)$$ **M1 A1**

$$(1 - tanh^2\,y)\frac{dy}{dx} = -sin\left(x + \frac{\pi}{4}\right)$$

Use $tanh\,y = cos\left(x + \frac{\pi}{4}\right)$

$$\left(1 - cos^2\left(x + \frac{\pi}{4}\right)\right)\frac{dy}{dx} = -sin\left(x + \frac{\pi}{4}\right)$$ **M1**

$$sin^2\left(x + \frac{\pi}{4}\right)\frac{dy}{dx} = -sin\left(x + \frac{\pi}{4}\right)$$

$$\frac{dy}{dx} = -\frac{1}{sin\left(x + \frac{\pi}{4}\right)} = -cosec\left(x + \frac{\pi}{4}\right)$$ **A1**

c)

First three terms of Maclaurin's series are $y(0)+y'(0)x+y''(0)\frac{x^2}{2!}$

$\frac{d^2y}{dx^2}=\frac{d}{dx}\left(-cosec\left(x+\frac{\pi}{4}\right)\right)=\frac{\cos\left(x+\frac{\pi}{4}\right)}{\sin^2\left(x+\frac{\pi}{4}\right)}$ **B1**

When x=0,

$y=tanh^{-1}\cos\left(x+\frac{\pi}{4}\right)=tanh^{-1}\frac{1}{\sqrt{2}}=\frac{1}{2}\ln\left(\frac{2+\sqrt{2}}{2-\sqrt{2}}\right)=\frac{1}{2}\ln\left(3+2\sqrt{2}\right)$ **M1**

$\frac{dy}{dx}=-\frac{1}{sin\left(x+\frac{\pi}{4}\right)}=-\frac{1}{\frac{1}{\sqrt{2}}}=-\sqrt{2}$

$\frac{d^2y}{dx^2}=\frac{\cos\left(x+\frac{\pi}{4}\right)}{\sin^2\left(x+\frac{\pi}{4}\right)}=\frac{\frac{1}{\sqrt{2}}}{\left(\frac{1}{\sqrt{2}}\right)^2}=\sqrt{2}$ **M1**

$y(0)+y'(0)x+y''(0)\frac{x^2}{2!}=\frac{1}{2}\ln\left(3+2\sqrt{2}\right)-\sqrt{2}x+\frac{\sqrt{2}x^2}{2}$ **M1 A1**

NO. 3

It is given that $y = 2^x$.

a. By differentiating $\ln y$ with respect to x, show that $\frac{dy}{dx} = 2^x \ln 2$. [3]

b. Write down $\frac{d^2y}{dx^2}$. [1]

c. Hence find the first three terms in the Maclaurin's series for 2^x. [3]

A level FURTHER MATHEMATICS 2020/Summer/Paper2_1/Q2

Mark scheme

Question	Answer	Marks
a	$\ln y = x \ln 2 \Rightarrow \frac{1}{y}\frac{dy}{dx} = \ln 2$	**M1 A1**
	$\frac{dy}{dx} = y \ln 2 = 2^x \ln 2$ **AG**	**A1**
		3
b	$\frac{d^2y}{dx^2} = 2^x (\ln 2)^2$	**B1**
		1
c	$y(0) = 1,\ y'(0) = \ln 2,\ y''(0) = (\ln 2)^2$	**B1**
	$2^x = y(0) + y'(0)x + \frac{y''(0)}{2!}x^2 + \ldots = 1 + (\ln 2)x + \frac{(\ln 2)^2}{2}x^2 + \ldots$	**M1 A1**
		3

Explanation

a.

$y = 2^x \rightarrow ln\, y = x\, ln\, 2$

Differentiate respect to x:

$\frac{1}{y}\frac{dy}{dx} = ln\, 2$ **M1 A1**

$\frac{dy}{dx} = y\, ln\, 2 = 2^x\, ln\, 2$ **A1**

b)

$\frac{d^2y}{dx^2} = \frac{d}{dx}(2^x \ln 2) = \ln 2 \frac{d(2^x)}{dx} = (\ln 2)^2 2^x$ **B1**

c)

First three terms of Maclaurin's series are $y(0) + y'(0)x + y''(0)\frac{x^2}{2!}$

When x=-0, y=1, $\frac{dy}{dx} = \ln 2, \frac{d^2y}{dx^2} = (\ln 2)^2$ **B1**

$y(0) + y'(0)x + y''(0)\frac{x^2}{2!} = 1 + x \ln 2 + \frac{(\ln 2)^2}{2}x^2$ **M1 A1**

NO. 4

Find the Maclaurin's series for $e^{1+x^2} + e^{1-x}$ up to and including the term in x^2 . [4]

A level FURTHER MATHEMATICS 2024/Summer/Paper2_1/Q2

Mark scheme

Question	Answer	Marks	Guidance
2	$e^{1+x^2} = e\left(e^{x^2}\right) = e\left(1 + x^2 + \ldots\right)$	B1	Changes e^{1+x^2} or e^{1-x} so that the formula given in the list of formulae (MF19) can be applied.
	$e^{1-x} = e\left(1 - x + \frac{1}{2}x^2 + \ldots\right)$	B1	
	$e\left(1 + x^2 + \ldots\right) + e\left(1 - x + \frac{1}{2}x^2 + \ldots\right)$	M1	Sums power series.
	$e^{1+x^2} + e^{1-x} = 2e - ex + \frac{3}{2}ex^2 + \ldots$	A1	
	Alternative method for question 2		
	$f'(x) = 2xe^{1+x^2} - e^{1-x}$	B1	Finds first derivative.
	$f''(x) = 4x^2e^{1+x^2} + 2e^{1+x^2} + e^{1-x}$	B1	Finds second derivative.
	$f(0) = 2e \quad f'(0) = -e \quad f''(0) = 3e$	M1	Evaluates their derivatives at zero.
	$e^{1+x^2} + e^{1-x} = 2e - ex + \frac{3}{2}ex^2 + \ldots$	A1	
		4	

Comment

Most candidates differentiated the function twice, then applied the general formula to find the Maclaurin's series. Among the few who attempted to use the existing Maclaurin's series given in the list of formulae, many did this incorrectly by replacing x in the series with $1 + x^2$.

NO. 5

It is given that $y = \sinh(x^2) + \cosh(x^2)$.

a. Use standard results from the list of formulae (MF19) to find the Maclaurin's series for y in terms of x up to and including the term in x^4. [2]

b. Deduce the value of $\frac{d^4y}{dx^4}$ when $x = 0$. [1]

c. Use your answer to part **(a)** to find an approximation to $\int_0^{\frac{1}{2}} y \, dx$, giving your answer as a rational fraction in its lowest terms. [2]

A level FURTHER MATHEMATICS 2021/Winter/Paper2_2/Q1

Mark scheme

Question	Answer	Marks	Guidance
a	$\sinh(x^2) + \cosh(x^2)$	M1	Combines using correct power series. Or sensible attempt at all four derivatives of y.
	$1+x^2+\frac{1}{2}x^4$	A1	
		2	
b	$\frac{1}{2}\times 4! = 12$	B1	
		1	
c	$\int_0^{\frac{1}{2}} 1+x^2+\frac{1}{2}x^4 \, dx$	M1	Substitutes their power series, must be at least $a + bx^2$.
	$=\left[x+\frac{1}{3}x^3+\frac{1}{10}x^5\right]_0^{\frac{1}{2}} = \frac{523}{960}$	A1	
		2	

Comment

a. This part was very well answered with most candidates accurately finding the Maclaurin's series using standard results from the list of formulae. Those who instead attempted to find the first four derivatives often made errors.

b. Candidates who had found the correct series in **part (a)** usually went on to state the correct value of the fourth derivative when $x = 0$. A few incorrectly multiplied their coefficient of x^4 by 1 or 2, rather than 24.

c. The majority of candidates integrated their answer to **part (a)** with most maintaining accuracy.

NO. 6

Find the Maclaurin's series for $e^x \tan x$ from first principles up to and including the term in x^2. [5]

A level FURTHER MATHEMATICS 2021/Winter/Paper2_1/Q1

Mark scheme

Question	Answer	Marks	Guidance
1	$e^x (\tan x + \sec^2 x)$	B1	Finds first derivative.
	$e^x (\tan x + 2\sec^2 x + 2\sec^2 x \tan x)$	M1 A1	Finds second derivative.
	$y(0) = 0$, $y'(0) = 1$, $y''(0) = 2$	M1	Evaluates derivatives at $x = 0$.
	$y = x + x^2$	A1	
		5	

Comment

This question was answered very well. Most candidates accurately differentiated twice to find the Maclaurin's series, though a few worked with an erroneous value of f'(0) without checking their work.

NO. 7

a. It is given that $y = \operatorname{sech}^{-1}(x+\frac{1}{2})$.

Express cosh y in terms of x and hence show that $\sinh y \frac{dy}{dx} = -\frac{1}{\left(x+\frac{1}{2}\right)^2}$. [3]

b. Find the first three terms in the Maclaurin's series for $\operatorname{sech}^{-1}(x+\frac{1}{2})$ in the form

$\ln a + bx + cx^2$,

where a, b and c are constants to be determined. [7]

A level FURTHER MATHEMATICS 2021/Summer/Paper2_1/Q7

Mark scheme

Question	Answer	Marks	Guidance
a	$\operatorname{sech} y = \dfrac{1}{\cosh y} = x + \frac{1}{2} \Rightarrow \cosh y = \left(x + \frac{1}{2}\right)^{-1}$	**B1**	Relates to cosh y,
	$\sinh y \dfrac{\mathrm{d}y}{\mathrm{d}x}$	**B1**	Differentiates LHS.
	$-\left(x + \frac{1}{2}\right)^{-2}$	**B1**	Differentiates RHS. AG.
		[3]	
b	$\sinh \dfrac{\mathrm{d}^2 y}{\mathrm{d}x^2} + \cosh y \left(\dfrac{\mathrm{d}y}{\mathrm{d}x}\right)^2 = 2\left(x + \dfrac{1}{2}\right)^{-3}$	**M1** **A1**	M1 A1 for LHS. B1 for RHS.
		B1	
	$y(0) = \operatorname{sech}^{-1}\left(\frac{1}{2}\right) = \cosh^{-1}(2) = \ln\left(2 + \sqrt{3}\right)$	**M1** **A1**	Relates to $\cosh^{-1}$ and uses logarithmic form.
	$y'(0) = -\dfrac{4}{\sqrt{3}} \quad y''(0) = \dfrac{16}{3\sqrt{3}}$	**M1**	Evaluates derivatives at $x = 0$.
	$y = \ln\left(2 + \sqrt{3}\right) - \dfrac{4}{\sqrt{3}}x + \dfrac{8}{3\sqrt{3}}x^2$	**A1**	
	Alternative method for question 7(b)		
	$\dfrac{\mathrm{d}y}{\mathrm{d}x} = -\dfrac{1}{\left(x + \frac{1}{2}\right)^2 \sqrt{\left(x + \frac{1}{2}\right)^{-2} - 1}} = -\dfrac{1}{\left(x + \frac{1}{2}\right)\sqrt{\frac{3}{4} - x - x^2}}$	**B1**	
	$\dfrac{\mathrm{d}^2 y}{\mathrm{d}x^2} = \frac{1}{2}\left(x + \frac{1}{2}\right)^{-1}\left(\frac{3}{4} - x - x^2\right)^{-\frac{3}{2}}(-1 - 2x) + \left(\frac{3}{4} - x - x^2\right)^{-\frac{1}{2}}\left(x + \frac{1}{2}\right)^{-2}$	**M1** **A1**	
	$y(0) = \operatorname{sech}^{-1}\left(\frac{1}{2}\right) = \cosh^{-1}(2) = \ln\left(2 + \sqrt{3}\right)$	**M1** **A1**	Relates to $\cosh^{-1}$ and uses logarithmic form.

	$y'(0) = -\frac{4}{\sqrt{3}} \quad y''(0) = \frac{16}{3\sqrt{3}}$	**M1**	Evaluates derivatives at x = 0.
	$y = \ln\left(2+\sqrt{3}\right) - \frac{4}{\sqrt{3}}x + \frac{8}{3\sqrt{3}}x^2$	**A1**	
		[7]	

Explanation

a)

$sech\, y = x + \frac{1}{2}$

$\frac{1}{\cosh y} = x + \frac{1}{2}$

$\cosh y = \frac{1}{x+\frac{1}{2}}$ **B1**

Differentiate respect to x

$\sinh y \cdot \frac{dy}{dx} = -\left(x+\frac{1}{2}\right)^{-2} = -\frac{1}{\left(x+\frac{1}{2}\right)^2}$ **B1 B1**

b)

differentiate respect to x again:

$\cosh y \left(\frac{dy}{dx}\right)^2 + \sinh y \cdot \frac{d^2y}{dx^2} = 2\left(x+\frac{1}{2}\right)^{-3}$ **M1 A1 B1**

When x=0, $y = \cosh^{-1}\frac{1}{\frac{1}{2}} = \ln\left(2+\sqrt{2^2-1}\right) = \ln\left(2+\sqrt{3}\right)$ **M1 A1**

(Use the result in formula sheet)

$\sin y = \frac{e^{\ln(2+\sqrt{3})} - e^{-\ln(2+\sqrt{3})}}{2} = \frac{(2+\sqrt{3})-(2-\sqrt{3})}{2} = \sqrt{3}$

Evaluate at 0 use the first derivative equation:

$$\sqrt{3}\frac{dy}{dx} = -\frac{1}{\left(0+\frac{1}{2}\right)^2}$$

$$\frac{dy}{dx} = -\frac{4}{\sqrt{3}}$$

Evaluate at 0 use the second derivative equation:

$$2\left(-\frac{4}{\sqrt{3}}\right)^2 + \sqrt{3}\cdot\frac{d^2y}{dx^2} = 2\left(0+\frac{1}{2}\right)^{-3}$$

$$\frac{32}{3} + \sqrt{3}\frac{d^2y}{dx^2} = 16$$

$$\frac{d^2y}{dx^2} = \frac{16}{3\sqrt{3}}$$ **M1**

Thus

$$y(0) + y'(0)x + y''(0)\frac{x^2}{2} = ln\left(2+\sqrt{3}\right) + \left(-\frac{4}{\sqrt{3}}\right)\cdot x + \frac{16}{\sqrt{3}}\cdot\frac{x^2}{2}$$

$$= ln\left(2+\sqrt{3}\right) - \frac{4}{\sqrt{3}}x + \frac{8}{\sqrt{3}}x^2$$ **A1**

Comment

a. The majority of candidates accurately differentiated both sides of the equation implicitly and showed enough working to justify the given answer.
b. Good candidates accurately used implicit differentiation again to find an equation involving the second derivative. For candidates who spotted that f(0) = $\cosh^{-1}2$, finding the logarithmic did not pose any difficulties. Better responses showed this connection and also maintained accuracy when substituting to find f′(0) and f′(0).